Lecture 11

The Jucys–Murphy spectrum

Objectives. *Lecture 10 gave each irreducible representation of S_n a Gelfand–Tsetlin basis, whose vectors are separated by their weights $\alpha = (a_1, \dots, a_n)$. To constrain these weights, we study s_i on the span of v and $s_i v$ using an invariant inner product, then use the Coxeter braid relation. Every weight satisfies three conditions defining a* content vector. *We show combinatorially that these are exactly the vectors $(c_1(T), \dots, c_n(T))$ of box contents of standard Young tableaux T. Lecture 12 proves that every content vector is a weight.*

By the end of this lecture you will be able to:

- For $1 \le i \le n-1$, use an invariant inner product to prove that a GZ vector v of weight α has $a_{i+1} \neq a_i$, that $s_i v = \pm v$ when $a_{i+1} = a_i \pm 1$, and that otherwise $s_i v - (a_{i+1} - a_i)^{-1} v$ is a GZ vector of weight $s_i \alpha$; compute s_i on the two vectors.
- Use the Coxeter relation $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ to show that no weight has three consecutive entries a, $a \pm 1$, a.
- Prove that every weight of S_n is a content vector, and decide whether a given vector is one.
- Prove that $T \mapsto (c_1(T), \dots, c_n(T))$ maps the standard Young tableaux with n boxes bijectively onto the content vectors of length n.

11.1 Two-dimensional analysis

Throughout, $V \in \mathrm{Irr}(S_n)$ has its lines V_γ (Theorem 10.6), and a GZ vector v has the weight $\alpha(v) = (a_1, \dots, a_n)$ of Definition 10.9. Fix on V an invariant inner product (Theorem 5.11(iii)), for which each $\rho(g)$ is unitary. The relations of Corollary 10.19 describe the action on the span of v and $s_i v$; we use the inner product to determine its structure.

Lemma 11.1 (GZ vectors and the inner product)**.**

(i) For the fixed invariant inner product, each s_i and each X_k acts on V by a self-adjoint operator, so the entries of every weight are real.

(ii) The lines V_γ are pairwise orthogonal.

(iii) If $w \in V$ is non-zero and $X_k w = b_k w$ for every k, then w is a GZ vector, of weight $(b_1, \dots, b_n)$.

Proof. (i) A transposition t has $t^2 = e$, so

$$\rho(t)^\dagger = \rho(t)^{-1} = \rho(t).$$

Each X_k is a sum of transpositions. Self-adjoint operators have real eigenvalues.

(ii) Two different paths have different weights (Corollary 10.15), so some X_k acts on their lines by different scalars. Eigenvectors of a self-adjoint operator for different eigenvalues are orthogonal.

(iii) Write $w = \sum_\gamma w_\gamma$ with $w_\gamma \in V_\gamma$. Each $(X_k - b_k)w_\gamma$ lies in V_γ, and the sum of the lines is direct, so $(X_k - b_k)w_\gamma = 0$ for every γ and k. So each γ with $w_\gamma \neq 0$ has weight $(b_1, \dots, b_n)$, and by Corollary 10.15 there is only one. ∎

Theorem 11.2 (The local structure). *Let v be a GZ vector of weight $\alpha = (a_1, \dots, a_n)$, and $1 \le i \le n-1$. Then $a_{i+1} \neq a_i$. Put $r = a_{i+1} - a_i$ and*

$$w = s_i v - r^{-1} v.$$

(i) If $r = \pm 1$, then $s_i v = rv$.

(ii) If $r \neq \pm 1$, then w is a GZ vector of weight

$$s_i\alpha = (a_1, \dots, a_{i-1}, a_{i+1}, a_i, a_{i+2}, \dots, a_n),$$

orthogonal to v for the fixed invariant inner product, with $\|w\|^2 = (1 - r^{-2})\,\|v\|^2$, and

$$s_i v = r^{-1} v + w, \qquad s_i w = (1 - r^{-2})\, v - r^{-1} w.$$

Proof. Recall from Corollary 10.19 that

$$X_i s_i v = a_{i+1} s_i v - v, \qquad X_{i+1} s_i v = a_i s_i v + v,$$
$$X_k s_i v = a_k s_i v \quad (1 \le k \le n,\ k \neq i, i+1).$$

$r \neq 0$. Suppose $a_{i+1} = a_i$, and call it a. By Corollary 10.19,

$$(X_i - a) s_i v = -v, \qquad (X_i - a) v = 0.$$

As a is real, $X_i - a$ is self-adjoint (Lemma 11.1(i)), and

$$\|v\|^2 = -\langle v, (X_i - a) s_i v\rangle = -\langle (X_i - a) v, s_i v\rangle = 0,$$

a contradiction.

A weight vector. By Corollary 10.19, and as $a_i + r = a_{i+1}$,

$$\begin{aligned} X_i w &= X_i s_i v - r^{-1} a_i v \\ &= a_{i+1} s_i v - v - r^{-1} a_i v \\ &= a_{i+1} s_i v - r^{-1} a_{i+1} v = a_{i+1} w. \end{aligned}$$

The second displayed relation gives

$$X_{i+1} w = a_i s_i v + v - r^{-1} a_{i+1} v = a_i w.$$

The third gives $X_k w = a_k w$ for $1 \le k \le n$, $k \neq i, i+1$. So if $w \neq 0$, it is a GZ vector of weight $s_i\alpha$ (Lemma 11.1(iii)). As $r \neq 0$, $s_i\alpha \neq \alpha$, and w is orthogonal to v (Lemma 11.1(ii)); this is also true if $w = 0$.

The norm. So $s_i v = r^{-1} v + w$ with w orthogonal to v, and s_i is unitary. The entries a_i, a_{i+1} are real, so r is real:

$$\|v\|^2 = \|s_i v\|^2 = r^{-2}\,\|v\|^2 + \|w\|^2\,.$$

So $\|w\|^2 = (1 - r^{-2})\,\|v\|^2$.

(i) If $r = \pm 1$, then $w = 0$, and $s_i v = r^{-1} v$, which is rv as $r^2 = 1$.
(ii) If $r \neq \pm 1$, then $w \neq 0$, and only $s_i w$ remains. As $s_i^2 = e$,

$$s_i w = v - r^{-1} s_i v = (1 - r^{-2})\, v - r^{-1} w. \qquad \blacksquare$$

Lecture 12 normalises v and w, so that the coefficient of $w/\|w\|$ in $s_i v$ becomes $\sqrt{1 - r^{-2}}$: Young's orthogonal form.

Example 11.3 (E again)**.** In § 10.6, $u = (1, 1, -2)$ and $f_1 = (1, -1, 0)$ are GZ vectors of E, of weights $(0, 1, -1)$ and $(0, -1, 1)$, and the inner product of $\mathbb{C}^3$ is invariant. For $i = 1$, r is 1 for u and -1 for f_1, and indeed

$$s_1 u = u, \qquad s_1 f_1 = -f_1.$$

For $i = 2$ and u, $r = -2$, and by Lecture 10's self-check

$$w = s_2 u + \frac{1}{2} u = \left(-\frac{1}{2} u + \frac{3}{2} f_1\right) + \frac{1}{2} u = \frac{3}{2} f_1,$$

a GZ vector of the swapped weight $(0, -1, 1)$, as the local theorem predicts. Here $\|f_1\|^2 = 2$ and $\|u\|^2 = 6$, so

$$\|w\|^2 = \frac{9}{4} \cdot 2 = \left(1 - \frac{1}{4}\right) \cdot 6 = (1 - r^{-2})\, \|u\|^2 .$$

Example 11.4 (Guard: the relations alone)**.** The local theorem's $r = \pm 1$ case used the inner product. The local operator relations alone do not suffice. On $\mathbb{C}^2$ with basis v, w, fix $a \in \mathbb{C}$ and let

	v	w
X_i	av	$(a+1)w$
X_{i+1}	$(a+1)v$	aw
s_i	$v + w$	$-w$

Then

$$s_i^2 = \mathbb{1}, \qquad s_i X_i + 1 = X_{i+1} s_i, \qquad X_i s_i + 1 = s_i X_{i+1}.$$

Here $a_{i+1} = a_i + 1$ for v, but $s_i v \neq v$. The line $\mathbb{C}w$ is invariant. An invariant line must be an eigenline $\mathbb{C}v$ or $\mathbb{C}w$ of X_i, and $s_i v \notin \mathbb{C}v$. Thus $\mathbb{C}w$ has no invariant complement. Suppose an inner product made X_i and s_i self-adjoint. Then a and $a + 1$ are real, and their eigenvectors v and w are orthogonal, so

$$\langle s_i v, w \rangle = \|w\|^2 , \qquad \langle v, s_i w \rangle = 0.$$

Self-adjointness of s_i would give $\|w\|^2 = 0$, a contradiction. Every complex representation of S_n has an invariant inner product (Theorem 5.11(iii)), for which these operators are self-adjoint (Lemma 11.1(i)). So the displayed operators cannot arise by restriction to an invariant subspace of such a representation.

11.2 The braid exclusion

The local theorem supplies moves between weights. The braid relation gives a further restriction on triples of neighbouring entries.

Definition 11.5 (Admissible transpositions). For $\alpha = (a_1, \dots, a_n) \in \mathbb{C}^n$ and $1 \le i \le n-1$, $s_i\alpha$ exchanges a_i and a_{i+1}. The transposition s_i is *admissible* for α if a_{i+1} is neither $a_i + 1$ nor $a_i - 1$.

By Theorem 11.2(ii), if α is the weight of a GZ vector of V and s_i is admissible for α, then $s_i\alpha$ is a weight of the same V.

Lemma 11.6 (The braid exclusion). *Let $n \ge 3$ and $1 \le i \le n-2$. No weight $\alpha \in \mathrm{Spec}(n)$ has*

$$a_{i+2} = a_i, \qquad a_{i+1} = a_i \pm 1.$$

Proof. Suppose a GZ vector v has $a_{i+1} = a_i + 1$ and $a_{i+2} = a_i$, so that $a_{i+2} = a_{i+1} - 1$. By Theorem 11.2(i),

$$s_i v = v, \qquad s_{i+1} v = -v.$$

So

$$s_i s_{i+1} s_i v = s_i s_{i+1} v = -s_i v = -v,$$

$$s_{i+1} s_i s_{i+1} v = -s_{i+1} s_i v = -s_{i+1} v = v.$$

The Coxeter braid relation of § 10.5 makes these equal, so $v = -v$, and $v = 0$, a contradiction. If $a_{i+1} = a_i - 1$, exchange the signs. ∎

Example 11.7 (The vector $(0,1,0)$). The vector $(0,1,0)$ passes every test of Theorem 11.2: no two neighbours are equal, and r is 1 for $i = 1$ and -1 for $i = 2$. A GZ vector of this weight would be fixed by s_1 and negated by s_2, which the braid relation forbids. So $(0,1,0) \notin \mathrm{Spec}(3)$.

Lemma 11.8 (Moving an entry). *Let α be the weight of a GZ vector of V, and $1 \le p \le q \le n$.*

(i) If $a_j \ne a_q \pm 1$ for $p \le j < q$, then moving the entry a_q to place p, and $a_p, \dots, a_{q-1}$ one place to the right, gives the weight of a GZ vector of V.

(ii) If $a_j \ne a_p \pm 1$ for $p < j \le q$, then moving a_p to place q, and $a_{p+1}, \dots, a_q$ one place to the left, does too.

Proof. (i) If $p = q$, no swaps are needed. Otherwise apply s_{q-1}, then s_{q-2}, and so on down to s_p. Each exchanges the moving entry a_q with its left neighbour a_j, $p \le j < q$. The hypothesis rules out a difference of ± 1. At each step the current tuple is a weight, so its neighbouring entries are also unequal by Theorem 11.2. Thus each swap gives another weight of V by the local theorem.

(ii) The same, with s_p, then s_{p+1}, up to s_{q-1}. ∎

Lemma 11.9 (Truncation).

(i) Every weight has $a_1 = 0$.

(ii) If $n \ge 2$ and $(a_1, \dots, a_n) \in \mathrm{Spec}(n)$, then $(a_1, \dots, a_{n-1}) \in \mathrm{Spec}(n-1)$.

(iii) The weights of S_1 and S_2 are

$$\mathrm{Spec}(1) = \{(0)\}, \qquad \mathrm{Spec}(2) = \{(0,1),\ (0,-1)\}.$$

Proof. (i) $X_1 = 0$.

(ii) Let v be a GZ vector of V, in the line of $\gamma = (U_1 \to \cdots \to U_n)$. Then v lies in $W_{n-1} \cong U_{n-1}$ (Theorem 10.6(i)). An isomorphism $\varphi : W_{n-1} \to U_{n-1}$ of S_{n-1}-representations commutes with $X_1, \dots, X_{n-1}$, which lie in $\mathbb{C}[S_{n-1}]$. So $X_k\varphi(v) = a_k\varphi(v)$ for $k \le n-1$, and $\varphi(v)$ is a GZ vector of U_{n-1} of weight $(a_1, \dots, a_{n-1})$, by Lemma 11.1(iii) for S_{n-1}.

(iii) S_1 has only $\mathbf{1}$. The irreducibles of S_2 are the lines $\mathbf{1}$ and sgn, and $X_2 = (1\,2)$ acts on them by 1 and -1. ∎

11.3 Spec(n) ⊆ Cont(n)

The initial-entry condition, truncation, admissible moves and the braid exclusion lead to the following three conditions. We will prove that every weight satisfies them.

Definition 11.10 (Content vectors). A vector $\alpha = (a_1, \dots, a_n) \in \mathbb{C}^n$ is a *content vector* if

(C1) $a_1 = 0$;

(C2) for $1 < q \le n$, $a_q - 1$ or $a_q + 1$ is one of $a_1, \dots, a_{q-1}$;

(C3) if $a_p = a_q = a$ with $p < q$, then $a - 1$ and $a + 1$ are both among $a_{p+1}, \dots, a_{q-1}$.

Cont(n) is the set of content vectors of length n.

The initial-entry condition (C1) and the earlier-neighbour condition (C2) make every entry of a content vector an integer. The conditions on a_q involve only $a_1, \dots, a_q$, so for $n \ge 2$ dropping the last entry of a content vector of length n gives one of length $n-1$. The name comes from § 11.4.

Example 11.11 (Cont(3)). By the earlier-neighbour condition (C2), $a_2 = \pm 1$. Let $a_2 = 1$. The same condition says $a_3 - 1$ or $a_3 + 1$ is 0 or 1, so

$$a_3 \in \{-1, 0, 1, 2\}.$$

The repeated-entry condition (C3) removes 1, as nothing lies between the two 1s, and 0, as -1 does not lie between the two 0s. The case $a_2 = -1$ is the mirror image, so

$$\mathrm{Cont}(3) = \{(0,1,2),\ (0,1,-1),\ (0,-1,1),\ (0,-1,-2)\}.$$

Theorem 11.12 (Weights are content vectors). *For every $n \ge 1$, Spec(n) ⊆ Cont(n). In particular, every weight is a vector of integers.*

Proof. By induction on n. The sets Spec(1) and Spec(2) of Lemma 11.9(iii) are Cont(1) and Cont(2), by (C1) and (C2). Let $n \ge 3$, and let $\alpha \in \mathrm{Spec}(n)$ be the weight of a GZ vector of V. By Lemma 11.9(ii) and induction, $(a_1, \dots, a_{n-1}) \in \mathrm{Cont}(n-1)$; so (C1) holds, and (C2) and (C3) hold for $q < n$. Let $a = a_n$.

(C2) for $q = n$. Suppose neither $a-1$ nor $a+1$ is among $a_1, \dots, a_{n-1}$. Then s_{n-1} is admissible for α, so $(a_1, \dots, a_{n-2}, a, a_{n-1}) \in \mathrm{Spec}(n)$, and by Lemma 11.9(ii) and induction

$$(a_1, \dots, a_{n-2}, a) \in \mathrm{Spec}(n-1) \subseteq \mathrm{Cont}(n-1).$$

Its last entry a stands in place $n - 1 \ge 2$, so by (C2) $a - 1$ or $a + 1$ is among $a_1, \dots, a_{n-2}$: a contradiction.

(C3) for $q = n$: the set-up. Let $a_p = a$ with $p < n$, and take p largest, so that a is not among $a_{p+1}, \dots, a_{n-1}$; a smaller p only lengthens the range in (C3). Suppose $a - 1$ is not among them either; the case of $a + 1$ is the same with the signs exchanged.

At most one $a+1$. If a_j and $a_{j'}$, $p < j < j' < n$, both equal $a+1$, then (C3) for $(a_1, \dots, a_{n-1})$ puts a between them, against the choice of p. So every a_j with $p < j < n$, except at most one equal to $a+1$, has $a_j \neq a$ and $a_j \neq a \pm 1$.

No $a+1$. Move a_n to place $p+1$ (Lemma 11.8(i)). The result is a weight with two equal neighbours a, a in places p and $p+1$, against Theorem 11.2.

One $a+1$*, at* j. Move a_p to place $j-1$ and then a_n to place $j+1$ (Lemma 11.8(ii), (i)). The result is a weight with the consecutive entries a, $a+1$, a, against Lemma 11.6. ∎

Lecture 12 proves the converse, $\mathrm{Cont}(n) \subseteq \mathrm{Spec}(n)$, by counting.

11.4 Content vectors and standard tableaux

Definition 11.13 (Young diagrams and standard tableaux)**.** Let $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash n$ (Definition 2.15). Write $\ell(\lambda) = \ell$ for its number of parts. Its *Young diagram* is the set of boxes $b = (i, j)$ with $1 \le i \le \ell$ and $1 \le j \le \lambda_i$, drawn the English way: rows from top to bottom, each starting in the first column. λ is its *shape*. The *content* of $b = (i, j)$ is $c(b) = j - i$; the boxes of one content form a *diagonal*. A *standard Young tableau* T of shape λ fills the boxes with $1, \dots, n$, each once, increasing along every row and down every column. $\mathrm{SYT}(\lambda)$ is the set of them, $\mathrm{SYT}(n)$ the union over $\lambda \vdash n$, and $c_k(T)$ is the content of the box holding k.

For instance, the tableau

$$T = \begin{array}{|c|c|c|}\hline 1 & 2 & 4 \\ \hline 3 & 5 \\ \cline{1-2}\end{array} \qquad \text{has} \qquad (c_1(T), \dots, c_5(T)) = (0, 1, -1, 2, 0).$$

Three facts, for T of shape λ:

(a) If a box b' is distinct from b and lies weakly above and to the left of b, the entry of T in b' is smaller: go along the row of b' to the column of b, then down. So the entries increase down each diagonal.

(b) The contents of the boxes of λ are the integers from $1 - \ell$ to $\lambda_1 - 1$: the first row and the first column give them all.

(c) An *addable* box lies outside the diagram and leaves a Young diagram when added. It is $(i, \lambda_i + 1)$ for some $i \le \ell + 1$, with $\lambda_{\ell+1} = 0$. Its content $\lambda_i + 1 - i$ strictly decreases with i, so different addable boxes have different contents.

Proposition 11.14 (Content vectors and tableaux)**.** *The map* $T \mapsto (c_1(T), \dots, c_n(T))$ *is a bijection from* $\mathrm{SYT}(n)$ *onto* $\mathrm{Cont}(n)$.

Proof. Into $\mathrm{Cont}(n)$. Let $a_k = c_k(T)$. (C1): 1 is in $(1, 1)$, since entries increase along rows and columns. (C2): if $k > 1$ is in $b = (i, j)$, then $(i, j-1)$ or $(i-1, j)$ is a box, with a smaller entry by increase towards the lower right, and content $a_k - 1$ or $a_k + 1$.

(C3): if $a_p = a_q$ with $p < q$, then p and q lie on one diagonal, q lower, in $b = (i, j)$ say, with $i, j \ge 2$. The boxes $(i-1, j)$ and $(i, j-1)$, of contents $a_q + 1$ and $a_q - 1$, hold entries between p and q, by the same increase.

One-to-one. By increase down diagonals, the diagonal of content h holds the k with $a_k = h$, in increasing order from the top. A diagonal of content h starts at $(1, h+1)$ if $h \ge 0$, and at $(1-h, 1)$ if $h < 0$. Its boxes run unbroken from that first box. The number of occurrences of h therefore determines the diagonal, and the vector places each k.

Onto. By induction on n; $n = 1$ is clear. Let $a \in \mathrm{Cont}(n)$, $n \geq 2$. By induction, $(a_1, \ldots, a_{n-1})$ is the vector of some $T' \in \mathrm{SYT}(\mu)$, $\mu \vdash n-1$. It suffices to put n into an addable box of μ of content $a = a_n$: as n is largest, the resulting tableau is standard.

If a is among $a_1, \ldots, a_{n-1}$, let p be its last place and $b = (i, j)$ the box of p, the lowest box on its increasing diagonal.

By (C3), the diagonal of content $a+1$ has a box with an entry larger than p. By increase towards the lower right, that box is not above b, so it is $(i+m, j+1+m)$ for some $m \geq 0$. A Young diagram contains every box above and to the left of one of its boxes, so $(i, j+1) \in \mu$. Likewise, the box on the diagonal of content $a-1$ gives $(i+1, j) \in \mu$.

The box $(i+1, j+1)$ is absent, since b was the lowest box of its diagonal. Its upper and left neighbours are present, so it is addable, of content a.

For example, take T' to be the tableau T above with 5 deleted; its shape is $(3, 1)$. This illustrates the repeated-content case: the previous box of content 0 is at $(1, 1)$, with 2 to its right and 3 below. The addable box $(2, 2)$ has content 0; putting 5 there restores T.

If not, then $a \neq 0$, by (C1). Let $a > 0$. By (C2), $a-1$ or $a+1$ is a content of μ.

The content interval of μ runs from $1 - \ell(\mu)$ to $\mu_1 - 1$. As $a > 0$ is not among them, $a \geq \mu_1$; so $a-1$ is one, and $a = \mu_1$.

The box $(1, \mu_1 + 1)$ is addable, of content a.

If $a < 0$, likewise $a = -\ell(\mu)$, and $(\ell(\mu)+1, 1)$ is addable. ∎

The proposition is combinatorial: it labels no GZ vector until Lecture 12 proves $\mathrm{Spec}(n) = \mathrm{Cont}(n)$.

11.5 Worked example

We find the weights of S_3. By Theorem 11.12 they lie in Cont(3), which has four elements (Example 11.11). S_3 has four paths of length 3, one each to $\mathbf{1}$ and sgn and two to E, with different weights (Corollary 10.15). So $\mathrm{Spec}(3) = \mathrm{Cont}(3)$.

On $\mathbf{1}$ each X_k acts by $k-1$, and on sgn by $1-k$: their weights $(0, 1, 2)$ and $(0, -1, -2)$ are the content vectors of the one-row and the one-column tableaux. E has the other two, $(0, 1, -1)$ for u and $(0, -1, 1)$ for f_1 (Example 11.3), the content vectors of

$$\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array} \qquad \text{and} \qquad \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}.$$

And $(0, 1, 0)$ is not a weight (Example 11.7).

Self-check. Which of $(0, 1, -1, 0)$ and $(0, 1, 0, -1)$ are content vectors, and of which tableaux? *Answer:* in $(0, 1, -1, 0)$, both 1 and -1 lie between the two 0s, so (C3) holds; (C1) and (C2) hold too. It is the content vector of

$$\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & 4 \\ \hline\end{array},$$

of shape $(2, 2)$, and a weight of $\rho_E \circ \pi$ in Example 10.11. There the edge $E \to \rho_E \circ \pi$ has label $0 - 0 = 0$, so the other path through E gives its other weight $(0, -1, 1, 0)$. In $(0, 1, 0, -1)$, -1 does not lie between the two 0s, so it fails (C3) and is not a weight.

Summary.

- *The main results:* with an invariant inner product, the local structure of s_i on a GZ vector of an irreducible S_n-representation (Theorem 11.2); the braid exclusion (Lemma 11.6); every weight is a content vector (Theorem 11.12); content vectors are the contents of standard tableaux (Proposition 11.14).

- *Handed on:* the local moves, $\mathrm{Spec}(n) \subseteq \mathrm{Cont}(n)$ and $\mathrm{Cont}(n) \leftrightarrow \mathrm{SYT}(n)$ to Lecture 12.
- *Next:* Lecture 12 proves $\mathrm{Spec}(n) = \mathrm{Cont}(n)$, classifies the irreducibles by partitions, and writes Young's orthogonal form.

References

Vershik and Okounkov, *A new approach to the representation theory of the symmetric groups. II* (arXiv:math/0503040): the algebra $H(2)$, pp. 14–15; Prop. 4.1, p. 16; Thm 5.1 and Lemma 5.2, pp. 17–19; Prop. 5.3, p. 20. Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Def. 3.1.8 and Thm 3.1.10, pp. 86–89; Prop. 3.3.3 and Lemmas 3.3.4–3.3.6, pp. 103–106.[1]

[1] They write $\mathrm{Tab}(\lambda)$ for $\mathrm{SYT}(\lambda)$ and $C(T)$ for the vector of T. Both sources argue for Theorem 11.2(i) in the span of v and $s_i v$ alone; Example 11.4 shows why this course uses the inner product there.